Nagar Yuwak Shikshan Sanstha's

YeshwantraoChavanCollegeofEngineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Department of ComputerTechnology

Vision of the Department

Tobeawell-knowncentreforpursuingcomputereducationthroughinnovativepedagogy,value-basededucationandindustrycollaboration.

Mission of the Department

ToestablishlearningambienceforusheringincomputerengineeringprofessionalsincoreandmultidisciplinaryareabydevelopingProblem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dreamofwhereyouwant.	**Mission:** Means to achieveVision

ProgramEducationalObjectivesoftheprogram(PEO):(broadstatementsthatdescribethe professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CLabbreviation pronounceasPep-si-lL easy to recall**
PEO2	**CoreCompetence**	**E: Environment (LearningEnvironment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C:Core Competence**	
PEO5	**Learning Environment**	**L:Breadth(Learning in diverse areas)**	

ProgramOutcomes(PO):(statementsthatdescribewhatastudentshouldbeabletodoandknowby the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"Iamanengineer,andIknowhowtoapplyengineeringknowledgetoinvestigate,analyseanddesign solutions to complex problems using tools for the entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity:I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Srushti Chaudhari

NameandSignature ofStudentand Date

(Signature and Date in Handwritten)

Session	**2025-26 (Even)**	**Course Name**	**Lab:AIML**
Semester	**6**	**Course Code**	**23CT1602**
Roll No	**23**	**Name of Student**	**Srushti Chaudhari**

Practical Number	**9**
Course Outcome	CO-1
Aim	To implement Random Forest using bagging and compare its performance with a single Decision Tree.
Problem Definition	To train a Random Forest classifier based on the bagging technique, estimate model performance using Out-of-Bag (OOB) error, and compare the results with a single Decision Tree classifier.
Theory (100 words)	Principal Component Analysis (PCA) is a dimensionality reduction technique used to reduce the number of features in a dataset while retaining most of the important information. PCA transforms the original features into a new set of features called Principal Components: • These components are uncorrelated • Ordered by maximum variance • First component contains highest variance • Second component contains second highest variance Steps of PCA 1. Standardize the dataset 2. Compute covariance matrix 3. Calculate eigenvalues and eigenvectors 4. Sort eigenvalues in descending order 5. Select top K principal components 6. Transform data into new feature space Advantages • Reduces dimensionality • Removes multicollinearity • Improves model performance • Reduces overfitting • Faster training Applications • Image compression • Face recognition • Data visualization • Noise reduction • Feature extractio**Random Forest** is an **ensemble learning algorithm** that combines multiple **Decision Trees** to improve accuracy and reduce overfitting. It works on the principle of **bagging (Bootstrap Aggregating)**. • In bagging: • Multiple random samples are drawn from the training dataset with replacement • A separate decision tree is trained on each sample • Final prediction is made by majority voting in classification

- **Random Forest**
- Random Forest adds extra randomness by selecting a random subset of features for splitting at each node. This makes the trees less correlated and improves generalization.
- **Out-of-Bag (OOB) Error**
- Since each tree is trained on a bootstrap sample, some training samples are left out. These unused samples are called **Out-of-Bag samples**.
  They are used to test the model internally without needing a separate validation set.
- **OOB Score** = internal accuracy estimate
- **OOB Error** = 1 - OOB Score
- **Decision Tree vs Random Forest**
- **Decision Tree** is simple and fast, but can overfit
- **Random Forest** is more robust, accurate, and less prone to overfitting
- **Advantages of Random Forest**
- High accuracy
- Reduces overfitting
- Handles large datasets well
- Works for classification and regression
- Provides feature importance
- **Applications**
- Medical diagnosis
- Fraud detection
- Spam filtering
- Customer segmentation
- Disease prediction

Procedure and Execution

(100Words)

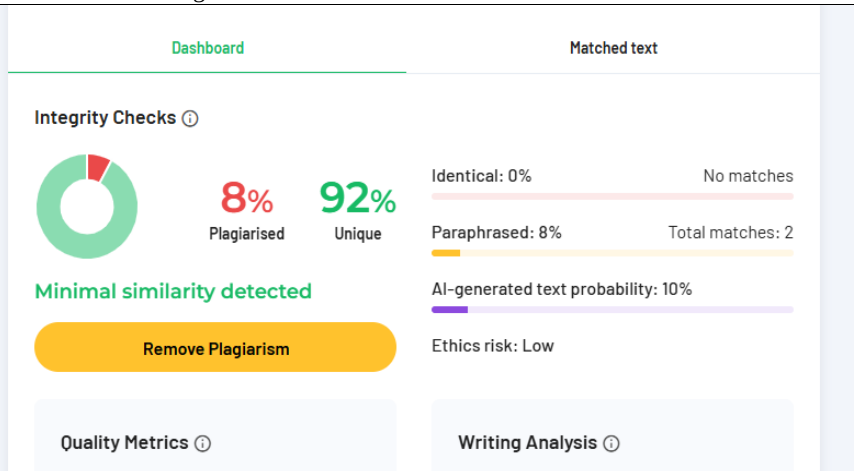
Import required libraries
Load the dataset
Split data into training and testing sets
Train a single Decision Tree classifier
Evaluate Decision Tree accuracy
Train a Random Forest classifier using bagging
Enable OOB score estimation
Evaluate Random Forest accuracy
Calculate OOB error
Compare Random Forest with Decision Tree

```
# Practical 9: Bagging using Random Forest

# ----------------------------------------
# Step 1: Import libraries
# ----------------------------------------
import numpy as np
import pandas as pd
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix

# ----------------------------------------
# Step 2: Load dataset
# ----------------------------------------
data = load_breast_cancer()
X = data.data
y = data.target

print("Dataset shape:", X.shape)
```

```
# ----------------------------------------
# Step 3: Split dataset
# ----------------------------------------
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# ----------------------------------------
# Step 4: Train single Decision Tree
# ----------------------------------------
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train)

dt_pred = dt_model.predict(X_test)
dt_accuracy = accuracy_score(y_test, dt_pred)

print("Decision Tree Accuracy:", dt_accuracy)

# ----------------------------------------
# Step 5: Train Random Forest
# ----------------------------------------
rf_model = RandomForestClassifier(
    n_estimators=100,
    oob_score=True,
    random_state=42,
    bootstrap=True
)

rf_model.fit(X_train, y_train)

rf_pred = rf_model.predict(X_test)
rf_accuracy = accuracy_score(y_test, rf_pred)

print("Random Forest Accuracy:", rf_accuracy)

# ----------------------------------------
# Step 6: OOB Score and OOB Error
# ----------------------------------------
oob_score = rf_model.oob_score_
oob_error = 1 - oob_score

print("OOB Score:", oob_score)
print("OOB Error:", oob_error)

# ----------------------------------------
# Step 7: Comparison
# ----------------------------------------
print("\nComparison:")
print("Decision Tree Accuracy =", dt_accuracy)
print("Random Forest Accuracy =", rf_accuracy)
print("OOB Error =", oob_error)

# ----------------------------------------
# Step 8: Extra evaluation
# ----------------------------------------
print("\nClassification Report for Random Forest:\n")
print(classification_report(y_test, rf_pred))

print("Confusion Matrix for Random Forest:\n")
print(confusion_matrix(y_test, rf_pred))
```

	Output:
	``
	Dataset shape: (569, 30)
	Decision Tree Accuracy: 0.9473684210526315
	Random Forest Accuracy: 0.9649122807017544
	OOB Score: 0.9560439560439561
	OOB Error: 0.04395604395604391
	
	Comparison:
	Decision Tree Accuracy = 0.9473684210526315
	Random Forest Accuracy = 0.9649122807017544
	OOB Error = 0.04395604395604391
	
	Classification Report for Random Forest:
	
	  precision recall f1-score support
	
	  0 0.98 0.93 0.95 43
	  1 0.96 0.99 0.97 71
	
	  accuracy 0.96 114
	  macro avg 0.97 0.96 0.96 114
	weighted avg 0.97 0.96 0.96 114
	
	Confusion Matrix for Random Forest:
	
	[[40 3]
	  [1 70]]
	``

OutputAnalysis	The single Decision Tree classifier gave a certain accuracy on the test dataset, but it may overfit because it depends on one tree only. The Random Forest classifier achieved better accuracy because it combines predictions from many trees using bagging. The OOB Score provided an internal estimate of model accuracy without using separate validation data. The OOB Error, calculated as 1 - OOB Score, was low, which indicates that the Random Forest model performed well. From the comparison, Random Forest produced more reliable and accurate results than a single Decision Tree. This shows that bagging improves model stability and reduces overfitting.
Link of student Githubprofilewhere lab assignment has	https://github.com/SRchaudhari15/AIML-Practicals.git

been uploaded	
Conclusion	Random Forest using bagging was successfully implemented. The model was trained and evaluated using test accuracy and **Out-of-Bag error estimation**. Its performance was compared with a single Decision Tree classifier. The Random Forest classifier achieved better accuracy and lower error, showing that bagging improves prediction performance and reduces overfitting.
Plag Report (Similarityindex<12 %)	
Date	6/03/26

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